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IITG SRIP 2026 Assessment: AI for Health

1. Understanding the Data and Visualization

The provided dataset contains 8 hours of polysomnography data per participant, consisting of Nasal Airflow (32 Hz), Thoracic Movement (32 Hz), and SpO□ (4 Hz). To effectively visualize this, a custom Python script was developed to align the varying sampling rates utilizing Pandas datetime indexing. The 8-hour recording was chunked into paginated 30-minute windows, allowing for clear clinical readability without signal compression. Annotated events (e.g., Apnea, Hypopnea) were parsed, standardized, and overlaid as shaded regions on the physiological signals using precise timestamp matching.

2. Signal Preprocessing and Dataset Creation

Raw respiratory signals contain significant noise, including baseline wander from patient movement and high-frequency artifacts. To isolate the relevant breathing frequency range (10 to 24 breaths per minute), a 4th-order Butterworth bandpass filter was applied with cutoffs at 0.17 Hz and 0.4 Hz.

Following filtering, the continuous temporal data was segmented into 30-second sliding windows with a 50% (15-second) overlap. To align the SpO□ data (4 Hz) with the respiratory signals (32 Hz) for deep learning, the SpO□ array was upsampled by a factor of 8. For labeling, each 30-second window was compared against the event annotations. If an abnormal breathing event overlapped the window for more than 50% of its duration (>15 seconds), the window inherited that event's label; otherwise, it was labeled as 'Normal'. The consolidated, structured dataset for all five participants was successfully exported as a Pickle object.

3. Basic Modeling and Evaluation

A 1D Convolutional Neural Network (1D CNN) was constructed to classify the labeled 30-second windows. The network ingests arrays of shape (960, 3) and utilizes stacked Conv1D and MaxPooling1D layers, followed by GlobalAveragePooling1D to extract hierarchical temporal features.

The evaluation was conducted using Leave-One-Participant-Out Cross-Validation (LOOCV). In each fold, the model was trained on four participants and validated on the completely unseen fifth participant, looping until all five participants served as the validation set.

4. Results and Discussion

The aggregated LOOCV results across all five participants are as follows:

- **Accuracy:** 91.37%
- **Precision:** 83.49%
- **Recall:** 91.37%

Confusion Matrix:

Plaintext

```
[[   0    0    0  163]  <-- Apnea
 [   0    0    0    3]  <-- Body Event
 [   0    0    0  593]  <-- Hypopnea
 [   0    0    0 8041]] <-- Normal
```

Discussion on Class Imbalance: While an accuracy of 91.37% appears high, the confusion matrix reveals a classic "Accuracy Paradox" driven by severe class imbalance. Because over 91% of the sleep windows represent normal breathing (8,041 out of 8,800 total windows), the baseline 1D CNN converged on a localized minimum: predicting 'Normal' for every single window. The model failed to detect the minority classes (Apnea and Hypopnea). Future iterations of this pipeline would require techniques such as SMOTE, focal loss, or strict class weighting during model compilation to penalize false negatives and force the network to learn the minority anomaly features.